

Executed planar rigid collision study

The defensible present recommendation for this repository is **coupled two-point normal contacts, warm-started global sequential impulse iteration, continuous collision detection, and adjustable collision-update time**. For the tested meter-scale polygon regime, the measured conservative setting is Box2D 2.4.1’s block solver with **eight primary steps and thirty-two velocity iterations** per 1/120 s output frame. This is a tested baseline, not a claim that an older release is the world’s best engine. Numerical adaptation is implemented, but the first controller does not earn a performance recommendation.

Scope and evidence

The method selection links the checked author documentation, explains why temporal/soft and block contacts differ, and records an independent restitution counterexample. The source lock pins both full source revisions and archive digests. Original sample sources, licenses and the old block solver source are retained unchanged.

The executable runner supports rotating convex polygons and compounds of convex polygons, persistent multipoint contacts, frictional stacks, many-body contact chains and CCD. A body’s fixtures stay attached rigidly; no elastic material deformation is being simulated. It does not solve a whole contact island’s complementarity equations exactly: two normal points in each manifold are solved together, then contacts are iterated globally.

The study ran 17 predeclared verification, training, test and stress scenes: hexagons, triangles, a concave L, spinning thin bars, thin-wall high-speed impact, stacks, 30:1 and 100:1 mass contrasts, a 12-body impact chain, a public friction sample adaptation and a 20-second tilted stack. The latter two are adaptations of published numerical samples, not reproductions of physical experiments. Every input is in `scenes.json`.

Mass, COM, inertia, fixture friction/restitution, initial conditions, gravity, duration and a common 0.01 m collision skin are preserved across comparisons. Mass properties use unrounded polygon cores. The collision skin is therefore a contact geometry tolerance; it contributes no extra mass. Density is areal kg/m², not a silently assumed 3D material density.

Acceptance rules

The declared full-history RMS tolerances are 0.02 m position, 0.05 m/s linear velocity and 0.05 rad/s spin. The normalized score is the largest ratio of a quantity’s error to its own tolerance; it must be at most one. Position and velocity use the Euclidean vector error before RMS averaging over time and bodies. Units are never mixed into an unscaled sum. Angle itself and worst-case per-body errors are not acceptance criteria in this first study.

For each backend and scene, primary steps are refined $4 \rightarrow 8 \rightarrow 16$ at 32 solver steps, and solver steps $8 \rightarrow 16 \rightarrow 32$ at 16 primary steps. The last two changes in both axes must each satisfy one quarter of the candidate error budget. The block backend’s solver steps are velocity iterations, with three position iterations fixed; temporal steps are actual substeps. Equal work counts across backends are not equal algorithms or costs. Spatial fixture discretization, collision skin and position iterations are not independently refined here.

Ten of seventeen block references pass these numerical consistency checks. Passing is not a convergence proof, and a converged solver can implement a physically unsuitable contact law. **The temporal reference passes its refinement check for symmetric rebound while failing the independent analytic result.** This is why fine simulation alone is insufficient evidence of authenticity.

Timing includes native integration and the actual controller’s feature/decision work. Shared state extraction, diagnostic extraction and JSON output are excluded; the adapter separately exposes subprocess end-to-end wall time. Each evaluated mode has an excluded warmup and five repetitions. Timings were grouped rather than randomized/interleaved and were collected on one shared Xeon Platinum 8573C Linux environment. They support a local engineering decision, not a portable throughput ranking or statistical confidence bound. First-study CSV preserves median/min/max; future runs additionally preserve every sample.

Results

Raw comparisons, reference checks and complete checksum-verified traces accompany this report. Counts below refer only to the ten qualified block references. Cross-backend agreement includes differences in stabilization and contact formulation; its reference favors a particular hard-contact approximation and must not be mistaken for experimentally established truth.

Mode	Collision updates / velocity iterations or temporal substeps	Cases within all RMS budgets
Block fast	1 / 1	4 / 10
Block standard	1 / 8	7 / 10
Block accurate	4 / 16	10 / 10
Block high	8 / 32	10 / 10
Block adaptive, frozen	variable	9 / 10
Temporal standard	1 / 4	6 / 10
Temporal accurate	4 / 16	7 / 10

The independent central rebound test uses equal unit-mass square bodies, velocities ± 2 m/s, restitution 0.6 and zero friction. Classical symmetry and the isolated restitution law give outgoing velocities ∓ 1.2 m/s and zero spin. The block accurate setting reproduces these within floating-point error. The measured temporal setting gives ∓ 1.056 m/s and opposite spins of 0.288 rad/s: 12% too little rebound speed and unwanted rotation. Linear momentum conservation alone would not detect this failure. The sweep records adverse as well as favorable solver refinements.

The adaptive block policy also gets those final rebound velocities and zero spin correct. Its one-update approach reaches the impact at a different output sample, however, yielding a normalized full-history error of 5.315. A correct impulse is not sufficient for accurate collision timing. In a discontinuous rigid impact, sampled velocity error is particularly sensitive to event timing; future evaluations should report impact-time error and post-impact observables separately alongside, rather than replacing, the declared trajectory score.

What adaptation actually achieved

The controller promotes fidelity from motion/size ratio, reported penetration, dynamic contact-island size and mass ratio; it demotes after a dwell period. It retains one native world and warm contact caches. All bodies share a chosen level, so isolated easy bodies can be refined unnecessarily. Known-contact features cannot certify safety before an unseen contact is detected.

Eight threshold/work candidates were calibrated before held-out evaluation. Only two training references qualified (flat slider and six-body stack). The triangle and 30:1 stack were excluded rather than silently used as truth. Several candidates had identical training errors and unexercised high-work branches. Runtime noise could decide between them; those training data do not identify a generally optimal controller.

All four held-out scenes with qualified references meet the frozen adaptive budget. **None is faster than the cheapest tested fixed block setting that meets that same budget.** These are measured per-scene median times, in ms:

Held-out qualified scene	Cheapest passing fixed setting	Fixed	Adaptive	Adaptive/fixed time
Concave L drop	Standard	0.727	1.101	1.51
Spinning thin bar	Accurate	3.433	11.072	3.23
Thin-wall CCD	Fast	0.049	0.578	11.80

Held-out qualified scene	Cheapest passing fixed setting	Fixed	Adaptive	Adaptive/fixed time
12-body impact chain	Accurate	8.242	22.066	2.68

A 20-repeat follow-up randomizes mode order within each round, preserving the original policy and settings. Adaptive/fixed median ratios are 1.53, 2.76, 12.39 and 2.51 respectively; this repeats the lack of speed benefit. All timing samples and round orders are retained, including end-to-end time. The shared machine is not CPU-isolated; some refinement work was concurrent. These are local repeated measurements, rather than formal confidence intervals.

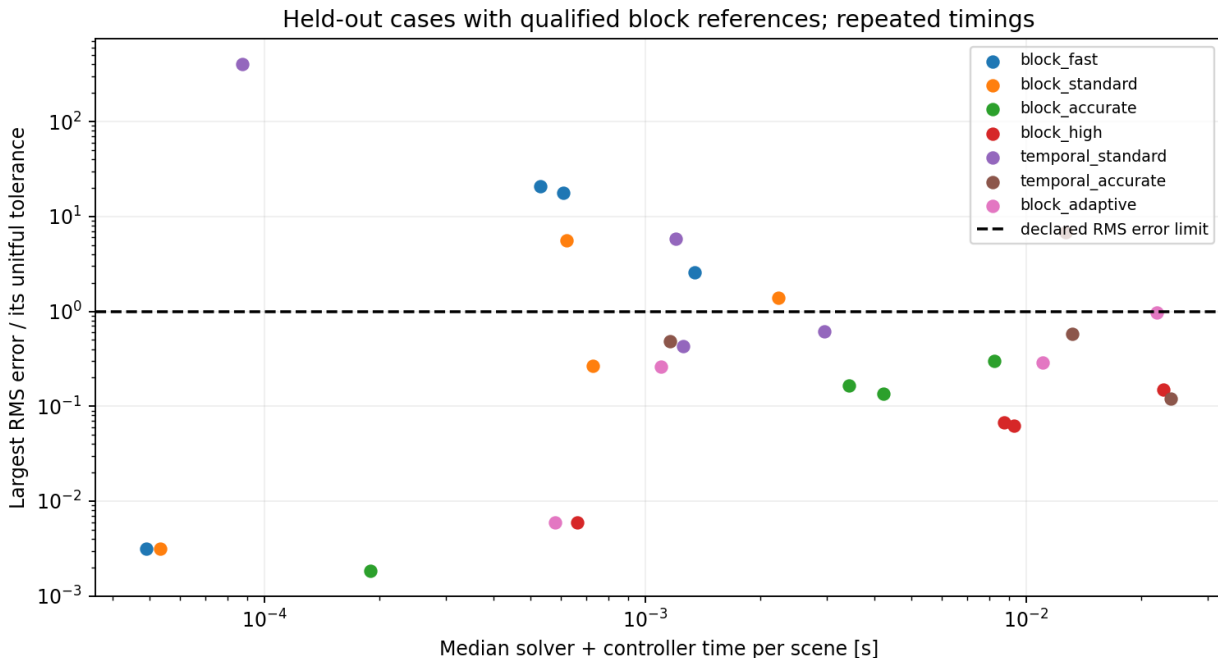


Figure 1: Accuracy and runtime on qualified held-out scenes

Adaptive code working is different from adaptive code being beneficial. Neither cheap tuning nor more work guarantees improvement. Keep fixed high as the current conservative choice, and expose cheap fixed settings explicitly for regimes whose outputs have been checked. The prototype is available for further research and should not be enabled as an automatic accuracy guarantee.

Hard cases and honest limits

Triangle/oblique drops, the twelve-body stack, both mass contrasts, friction slopes and the long tilted stack fail the initial block reference qualification. Their curves and timings remain in the data, but they cannot support favorable matched-error claims. Pointwise late trajectories of unstable stacks can diverge while aggregate quantities remain close; both must be reported explicitly. The exploratory follow-up refines primary steps $16 \rightarrow 32 \rightarrow 64$ at 128 velocity iterations, and iterations $32 \rightarrow 64 \rightarrow 128$ at 64 primary steps, still holding position iterations at three. It resolves the triangle drop, twelve-body stack and 30:1 contrast under the same reference budgets. Oblique hexagon, 100:1 contrast, public friction slopes and the long tilted stack remain unresolved. These checks do not retune the policy or revise the original held-out score. The follow-up records include late COM height, kinetic energy and momentum/energy diagnostics at each level. For example, the

oblique hexagon has late mean height near 0.53961 m and near-zero kinetic energy across refinements while its spin trajectory still misses the reference budget. Settled height alone cannot validate the impact history. The floor is finite; objects leaving it in long runs can legitimately fall below it, so negative late heights are not automatically penetration failures.

Comparing preserved first-study histories with the newly qualified 64×128 references shows why the default needs the higher preset. These are exploratory comparisons, not a replacement held-out evaluation:

Newly qualified scene	4×16 normalized error	8×32 normalized error
Triangle drop	1.630	0.130
Twelve-body stack	0.446	0.370
30:1 mass contrast	0.576	0.355

The 8×32 preset passes all ten initial qualified references and these three additional qualified scenes. The 4×16 preset passes twelve of those thirteen; it is a useful cheaper option, but cannot support the same tested accuracy scope. The adapter now defaults to 8×32 . The higher setting still has no accuracy guarantee in the four unresolved cases or unseen physical conditions. All follow-up comparisons are retained; no physical coefficients or controller thresholds were fitted to make them pass.

The benchmark does not include joints, 3D contact cones, materials with true elastic contact history, separate static/dynamic coefficients, rolling in the block backend, deformable FEM, GPU workloads or thousands of bodies. It cannot establish superiority over MuJoCo, Drake, Chrono, other rigid solvers or IPC/FEM. Our custom compliant local model is useful for contact-history research but is not an independent moving-polygon reference for these scenes.

Physical authenticity and the next admissible improvement

Current friction and restitution values are declared numerical inputs. Public source provenance authenticates where a value came from; it does not authenticate how a real surface behaves. To claim physical accuracy, obtain measurements with uncertainty for several impact speeds, angles and locations, including tangential rebound/spin and frictional stopping. Fit one transferable material-pair model on training observations and validate unseen conditions. Do not fit restitution or damping to compensate for a poorly resolved solver.

Keep the contact law fixed while adapting numerical work first. A next controller should predict approaching contacts, budget collision-time error, use measured solver residuals or cheap/high local comparisons, and allocate effort per independent contact island. Its overhead and missed promotions must be measured. Use the current held-out scenes as development evidence now; a revised controller needs new untouched test scenes before making a new generalization claim.

A later rigid/compliant/deformable switch must additionally preserve momentum, angular momentum, recoverable contact/material energy and internal history. Partitioning a body into spring-connected rigid cells adds material parameters and dispersion/objectivity concerns. The earlier rod experiment does not yet justify replacing continuum dynamics for arbitrary impact. No new friction law or publication-worthy adaptive superiority has been established by this study.

Sources supporting the method choice

- Erin Catto, Solver2D (2024): author discussion of iterative contacts, temporal substeps, softness and relaxation. Read for algorithm rationale, not as a performance result for our engine.
- Box2D simulation documentation: author recommendation of substeps and explanation of CCD, manifolds and contact settings. We tested the pinned implementations rather than assuming the recommended setting passed our cases.
- Pinned two-point block solver source: directly inspect the coupled normal complementarity solve. Coupling is existing prior art, not our invention.

- MuJoCo computation documentation: hard complementarity and convex soft friction are different model choices. Read for the limits of treating numerical convergence as physical truth; no MuJoCo throughput comparison was executed here.
- Anitescu and Potra (1997), Formulating Dynamic Multi-Rigid-Body Contact Problems with Friction as Solvable Linear Complementarity Problems: prior-art lead for global frictional contact formulations, checked in the earlier review. Its existence supports the lack of novelty in simply assembling coupled impulses; it does not establish our solver’s performance.

Reproduction

Build both pinned backends following the backend guide, then run:

```
python -m research.run_rigid_study --repeats 5 --output /tmp/new-rigid-study
python -m research.audit_rigid_study --pack --directory /tmp/new-rigid-study
python -m research.audit_rigid_study
python -m research.refine_rigid_references
python -m research.time_rigid_modes
```

A typeset PDF version accompanies this report. Rebuild it with Pandoc and a LaTeX installation:

```
pandoc research/rigid-study-report.md --resource-path=research --include-in-header=research/rigid-study-
```

The original frozen policy and histories stay in the repository. Repetition changes runtime samples and may change calibration ties; numerical results are compared with tolerances, not an unsupported bitwise determinism promise.